

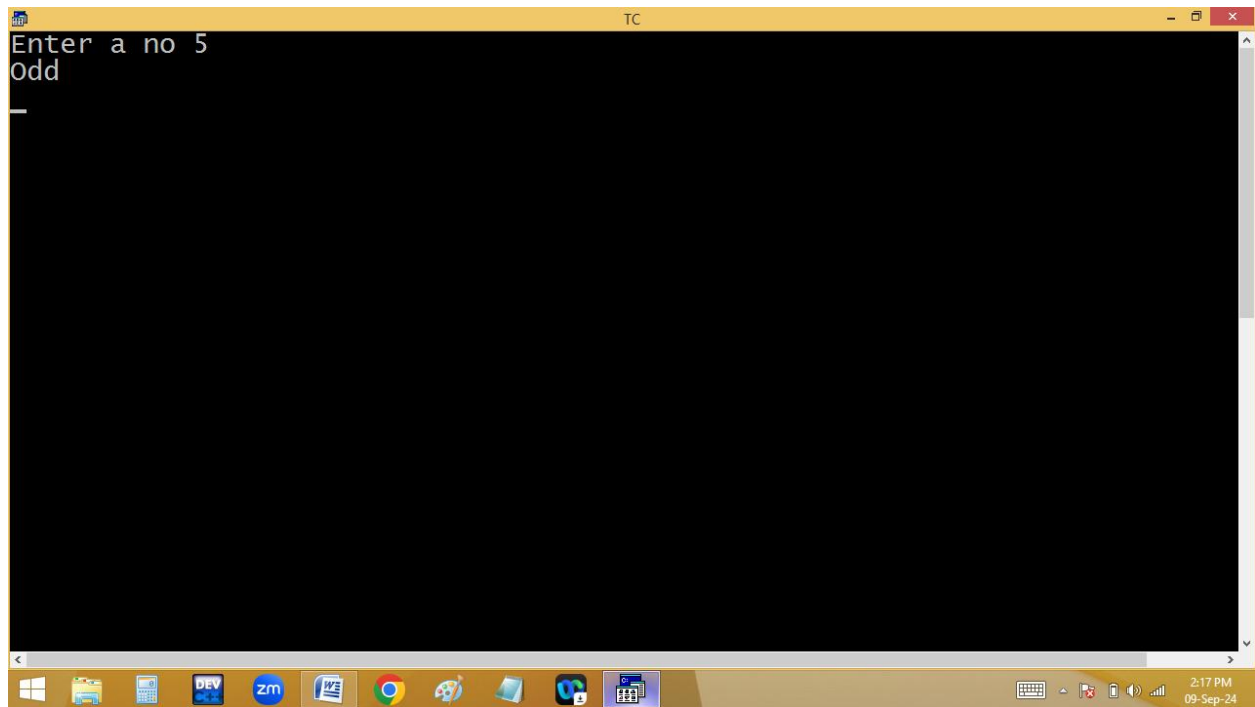
Finding even/odd using switch:

The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays the source code for a C program named E:2PM.C. The code is as follows:

```
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 2 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter a no ");scanf("%d",&n);
switch(n%2)
{
case 0: puts("Even");break;
default: puts("Odd");
}
getch();
}
```

The bottom window shows the execution of the program. It displays the prompt "Enter a no" followed by the user input "4", and then the output "Even".

Enter a no 4
Even



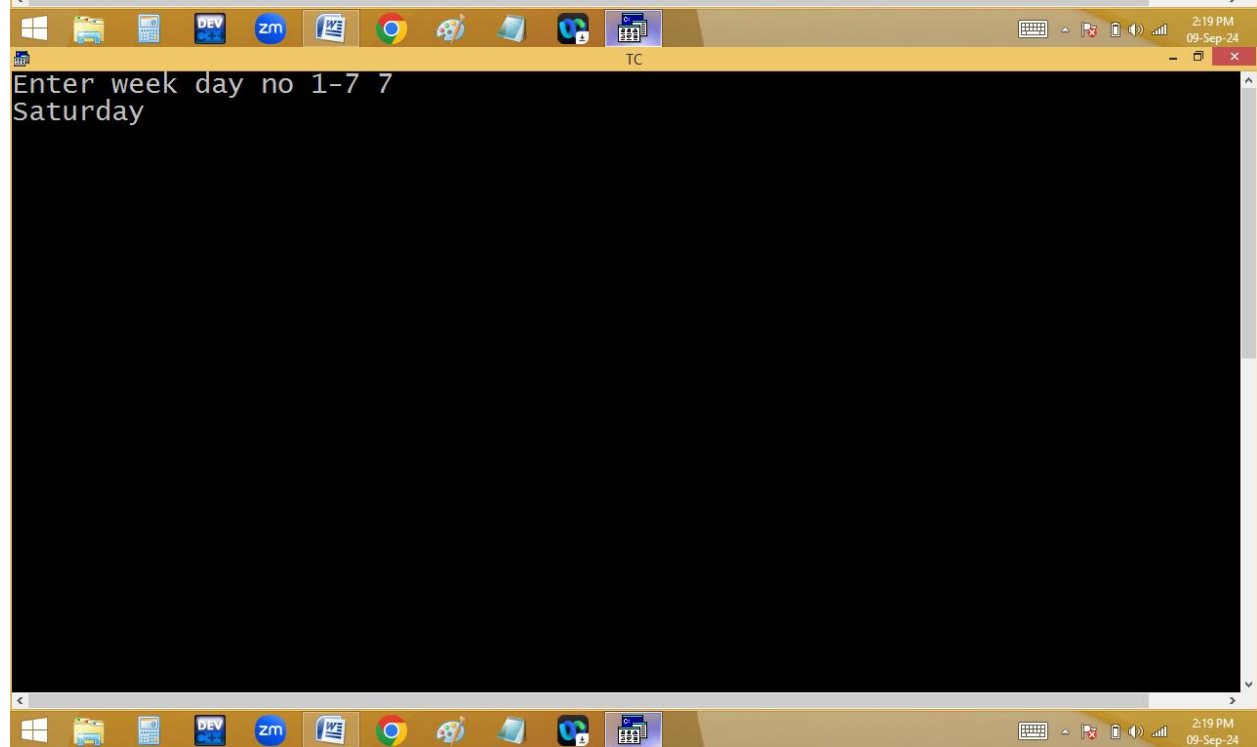
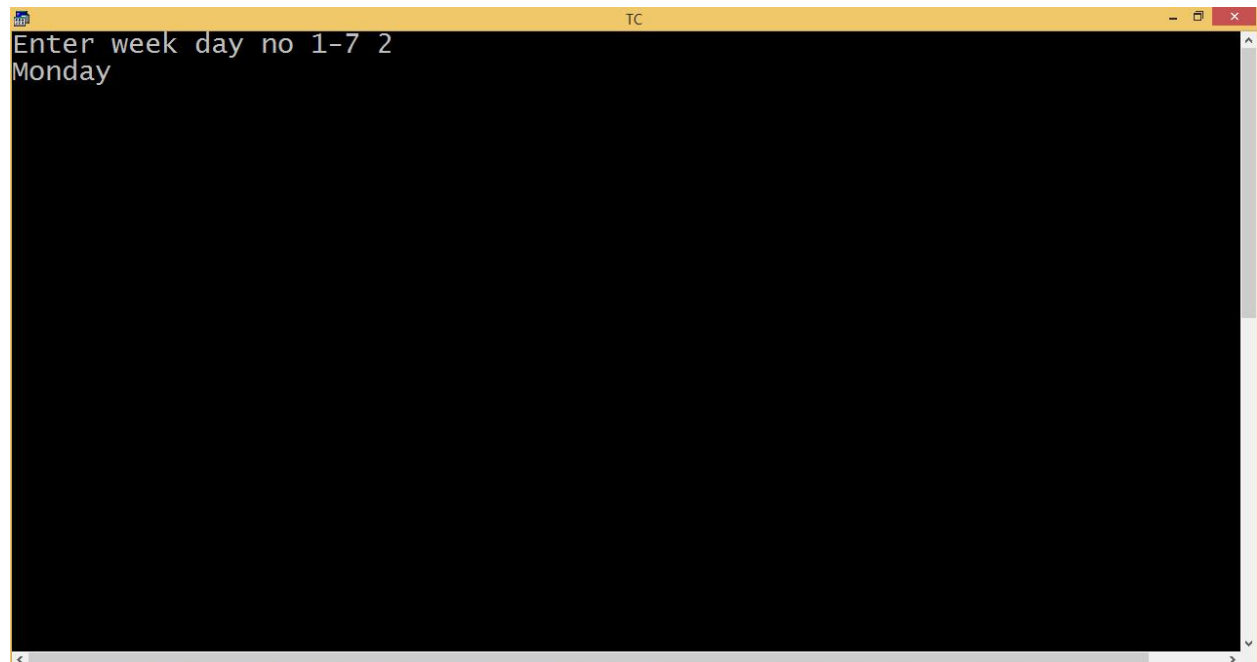
Finding weekday name:

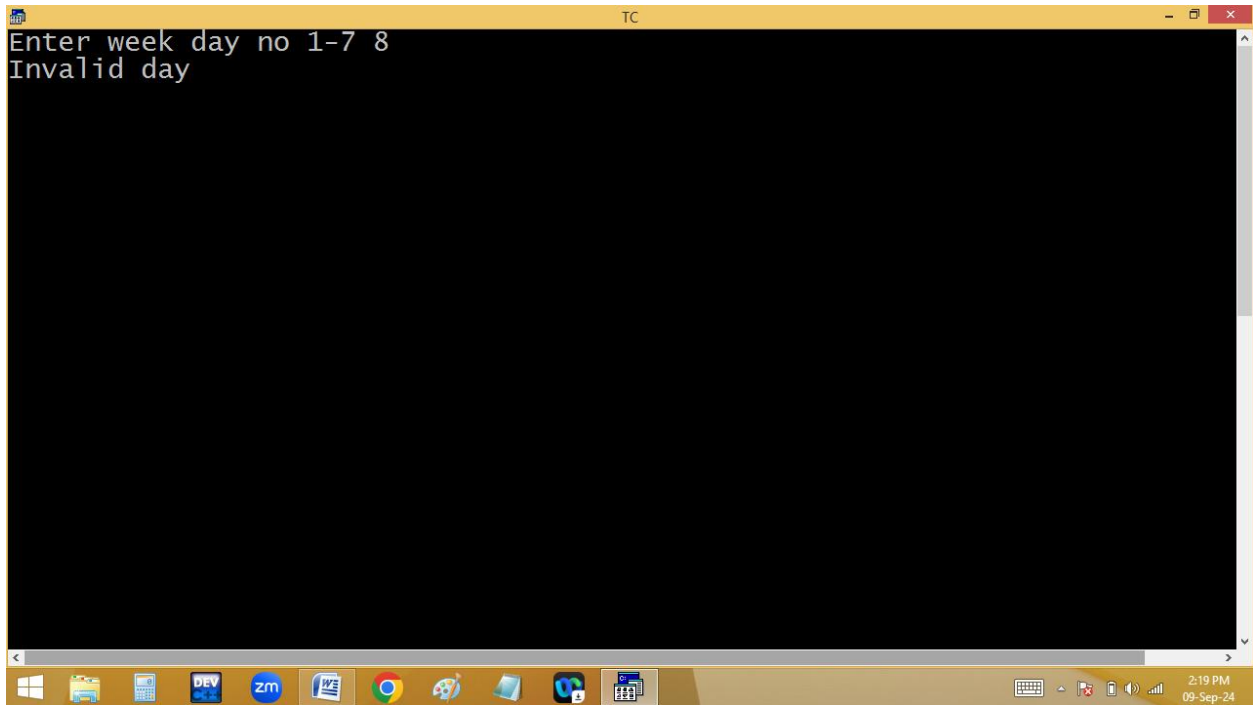
The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays the source code for a program named E:2PM.C. The code includes headers for stdio.h and conio.h, and defines a main function. Inside main, it declares an integer n, clears the screen with clrscr(), and prompts the user to enter a week day number (1-7). It uses a switch statement to map the input to the corresponding day of the week: 1 for Sunday, 2 for Monday, 7 for Saturday, and a default for 'Invalid day'. The program then gets the user's input and prints the result.

```
File Edit Run Compile Project Options Debug Break/watch
Line 14 Col 27 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter week day no 1-7 ");scanf("%d",&n);
switch(n)
{
case 1: puts("Sunday");break;
case 2: puts("Monday");break;
case 7: puts("Saturday");break;
default: puts("Invalid day");
}
getch();
}
```

The bottom window shows the execution of the program. It displays the prompt "Enter week day no 1-7" followed by the user input "1". The program then outputs "Sunday".

```
Enter week day no 1-7 1
Sunday
```





Hotel bill generation:

```
#include<stdio.h>

#include<conio.h>

void main()

{

int qty,op;

float amt=0;

start:

clrscr();

puts("\t\t\t HOTEL SAIRAM");

puts("\t\t\t AMEERPET - HYD");
```

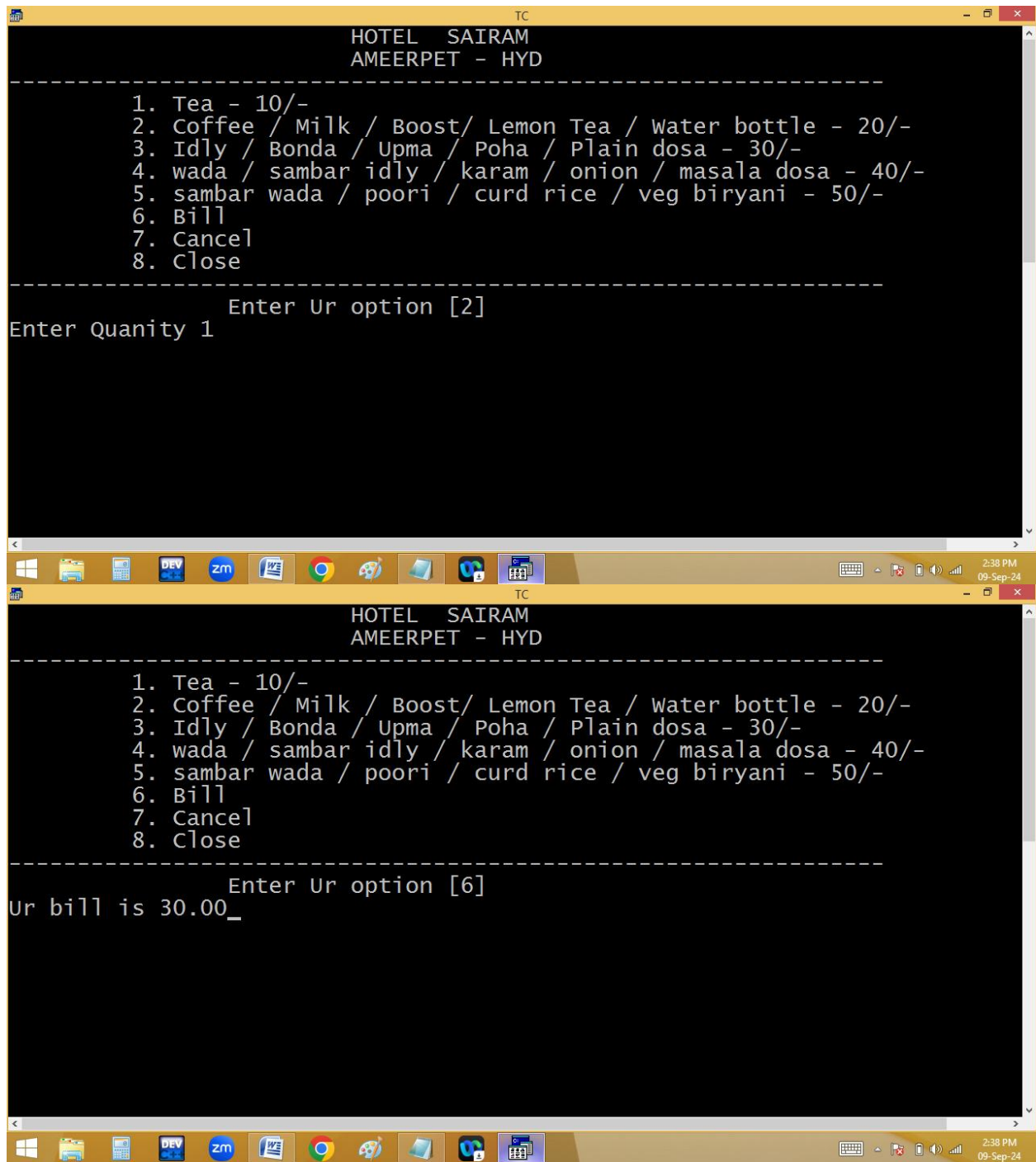
```
puts("-----");  
puts("\t 1. Tea - 10/-");  
puts("\t 2. Coffee / Milk / Boost/ Lemon Tea / Water bottle -  
20/-");  
puts("\t 3. Idly / Bonda / Upma / Poha / Plain dosa - 30/-");  
puts("\t 4. wada / sambar idly / karam / onion / masala dosa -  
40/- ");  
puts("\t 5. sambar wada / poori / curd rice / veg biryani - 50/-");  
puts("\t 6. Bill");  
puts("\t 7. Cancel");  
puts("\t 8. Close");  
puts("-----");  
printf("\t\tEnter Ur option [ ]\b\b") ;  
scanf("%d",&op);  
if(op<=5) {printf("Enter Quanity "); scanf("%d",&qty);}  
switch(op)  
{  
case 1: amt+=qty*10;break;  
case 2: amt+=qty*20;break;
```

```
case 3: amt+=qty*30;break;
case 4: amt+=qty*40;break;
case 5: amt+=qty*50;break;
case 6: printf("Ur bill is %.2f",amt); amt=0;getch(); break;
case 7: printf("Ur order cancelled",amt=0);getch();break;
case 8: return;
}
goto start;
}
```



```
TC
HOTEL SAIRAM
AMEERPET - HYD
-----
1. Tea - 10/-
2. Coffee / Milk / Boost/ Lemon Tea / Water bottle - 20/-
3. Idly / Bonda / Upma / Poha / Plain dosa - 30/-
4. wada / sambar idly / karam / onion / masala dosa - 40/-
5. sambar wada / poori / curd rice / veg biryani - 50/-
6. Bill
7. Cancel
8. Close
-----
Enter Ur option [8]
```

```
TC
HOTEL SAIRAM
AMEERPET - HYD
-----
1. Tea - 10/-
2. Coffee / Milk / Boost/ Lemon Tea / Water bottle - 20/-
3. Idly / Bonda / Upma / Poha / Plain dosa - 30/-
4. wada / sambar idly / karam / onion / masala dosa - 40/-
5. sambar wada / poori / curd rice / veg biryani - 50/-
6. Bill
7. Cancel
8. Close
-----
Enter Ur option [1]
Enter Quantity 1_
```



Read two numbers and perform arithmetic operation using menu:

Enter two numbers 3 6
M E N U
+. Add -. Sub *. Mul %. Mod /. Div E. Exit
Enter ur option[+] Sum = 9

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
#include<math.h>
```

```
void main()
```

```
{
```

```
float a,b;
```

```
char op;
```

```
start:
```

```
clrscr();
```

```
puts("*****  
*****");
```

```

printf("Enter two numbers "); scanf("%f %f",&a, &b);

puts("*****");

puts("\t\t\t M E N U");

puts("*****");

puts("\t\t\t +. Add");

puts("\t\t\t -. Sub");

puts("\t\t\t *. Mul");

puts("\t\t\t %. Mod");

puts("\t\t\t /. Div");

puts("\t\t\t E. Exit");

puts("*****");

printf("\t\t\tEnter Ur option[ ]\b\b");scanf(" %c",&op);

gotoxy(60,17); /* 60-col, 17-row */

switch(op)

{

case '+': printf("Sum=%.2f",a+b);break;

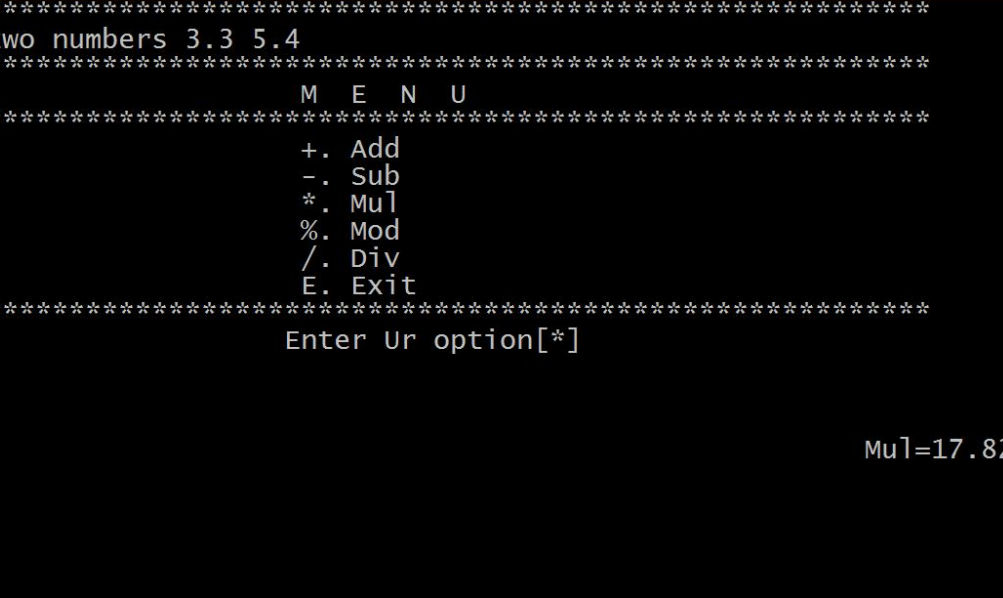
```

```
case '-': printf("Sub=%.2f",a-b);break;
case '*': printf("Mul=%.2f",a*b);break;
case '%': printf("Mod=%.2f",fmod(a,b));break;
case '/': printf("Div=%.2f",a/b);break;
case 'e': case 'E': return;
}
getch();
goto start;
}
```

```

TC
*****
Enter two numbers 2 3
*****
                M  E  N  U
*****
                +. Add
                -. Sub
                *. Mul
                %. Mod
                /. Div
                E. Exit
*****
                Enter Ur option[E]

```



```
*****
Enter two numbers 3.3 5.4
*****
                M E N U
*****
+. Add
-. Sub
*. Mu
%. Mod
/.. Div
E. Exit
*****
                Enter Ur option[*]

Mu=17.82_
```

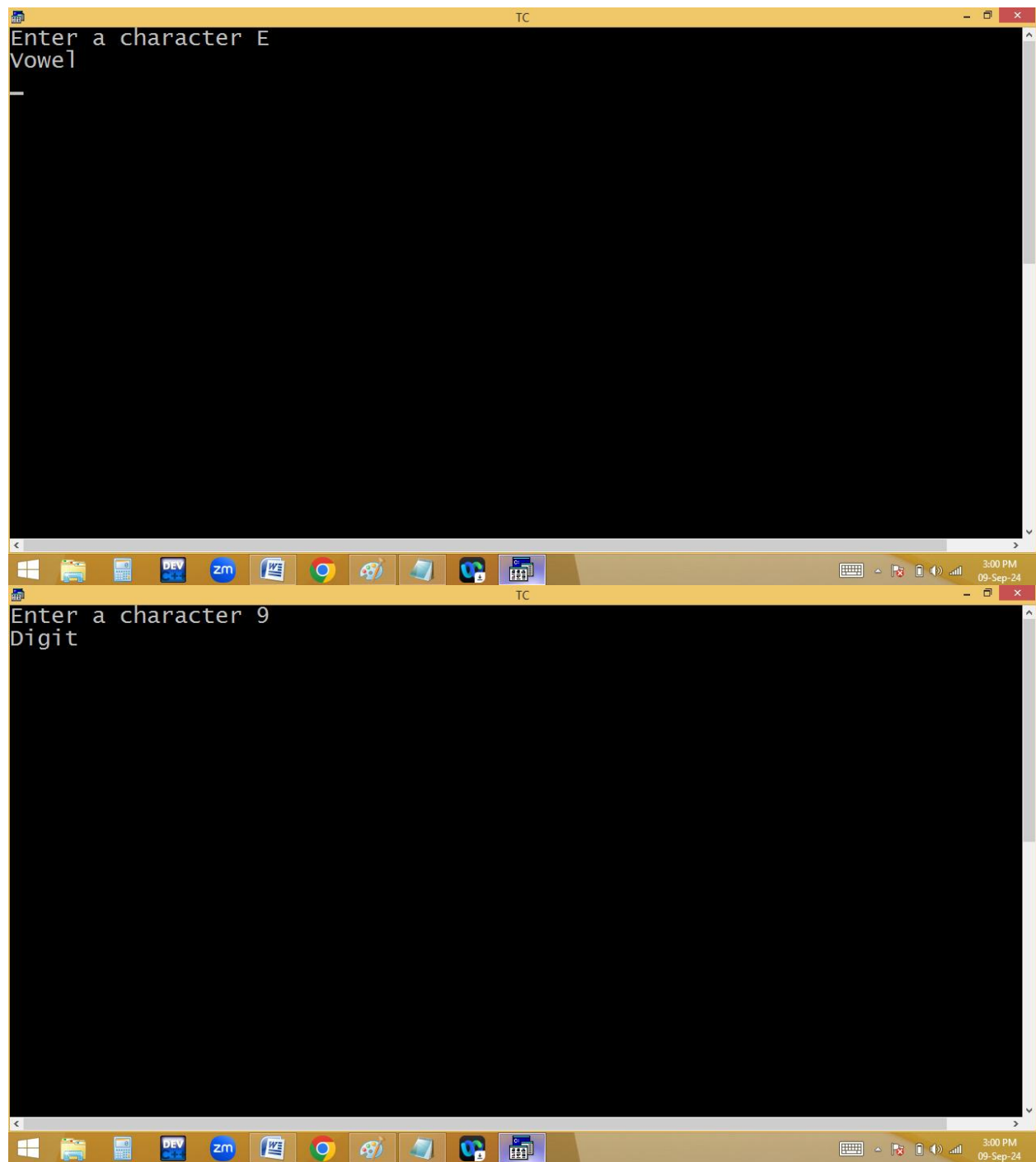
```
TC
*****
Enter two numbers 3 2
*****
          M E N U
*****
          +. Add
          -. Sub
          *. Mul
          %. Mod
          /. Div
          E. Exit
*****
          Enter Ur option[-]

sub=1.00_
```

Finding vowel/consonant using switch:

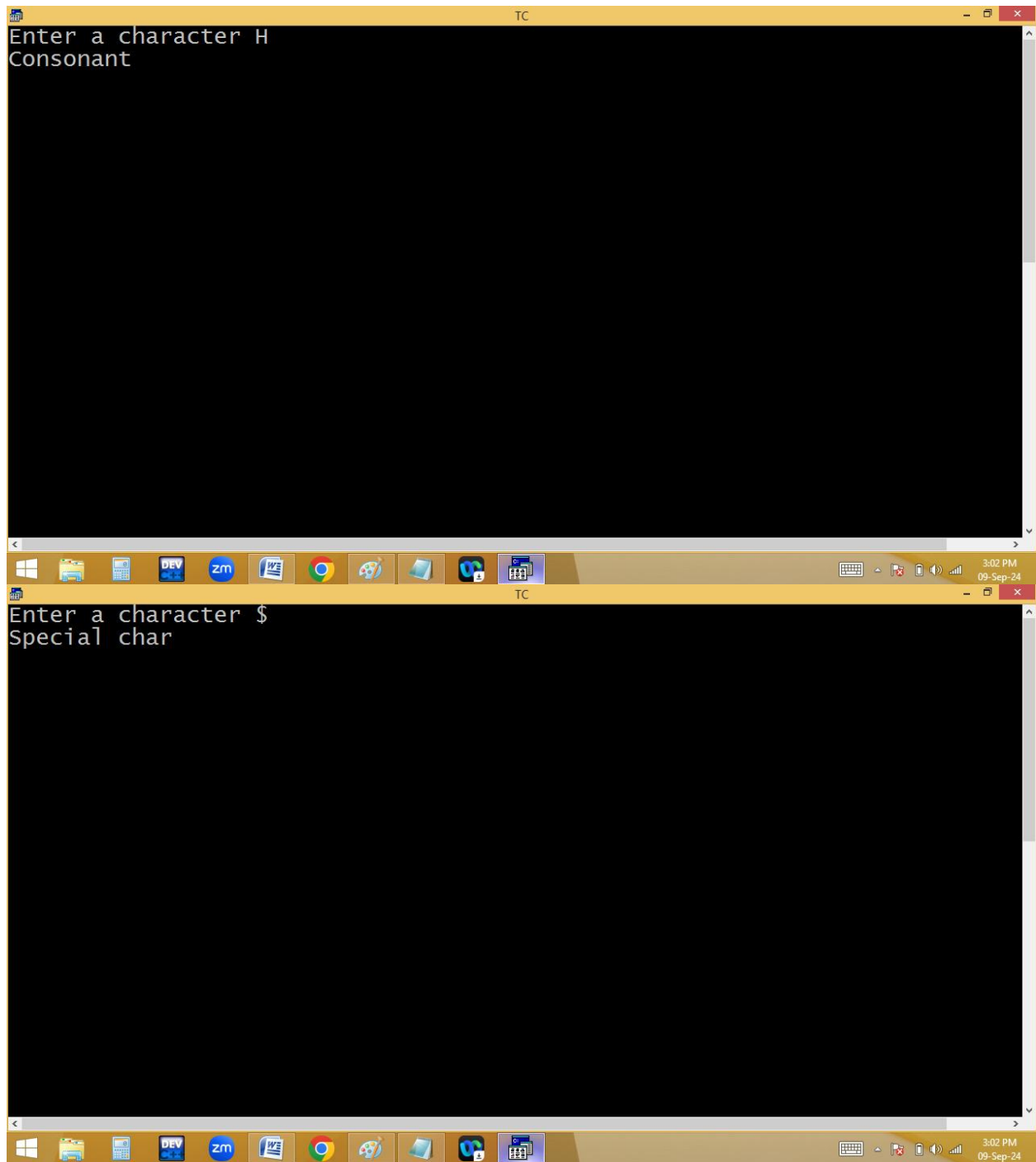
```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 19 Col 27 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter a character ");scanf("%c",&ch);
if(ch>='A'&&ch<='Z')ch+=32;
if(ch>='a' && ch<='z')
{
switch(ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u')
{
case 1: puts("Vowel");break;
default: puts("Consonant");break;
}
}
else if(ch>='0'&&ch<='9')puts("Digit");
else puts("Special char");
getch();
}

TC
Enter a character h
Consonant
```

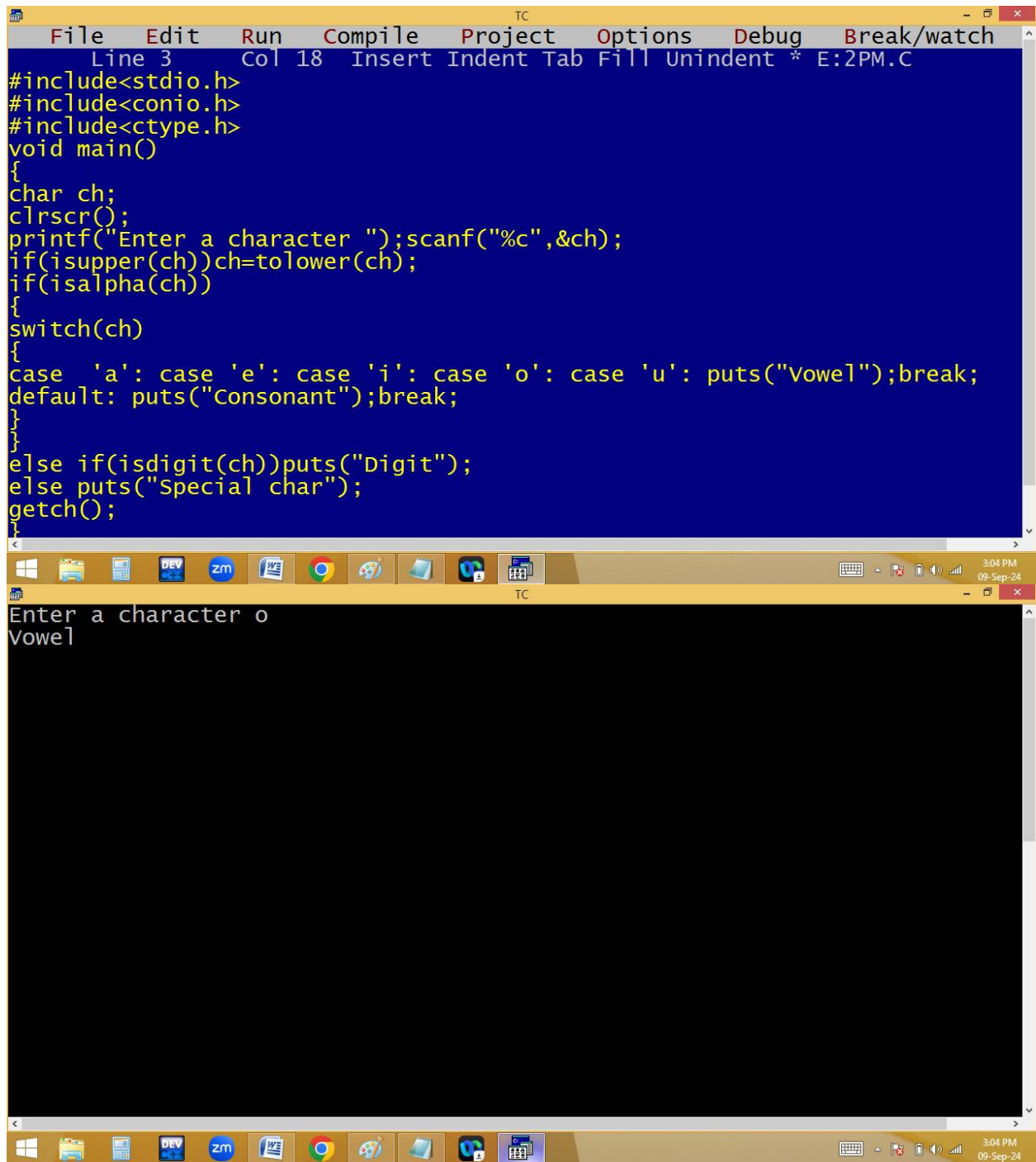



```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 51 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter a character ");scanf("%c",&ch);
if(ch>='A'&&ch<='Z')ch+=32;
if(ch>='a' && ch<='z')
{
switch(ch)
{
case 'a': case 'e': case 'i': case 'o': case 'u':_puts("Vowel");break;
default: puts("Consonant");break;
}
}
else if(ch>='0'&&ch<='9')puts("Digit");
else puts("Special char");
getch();
}
```

Enter a character I
Vowel

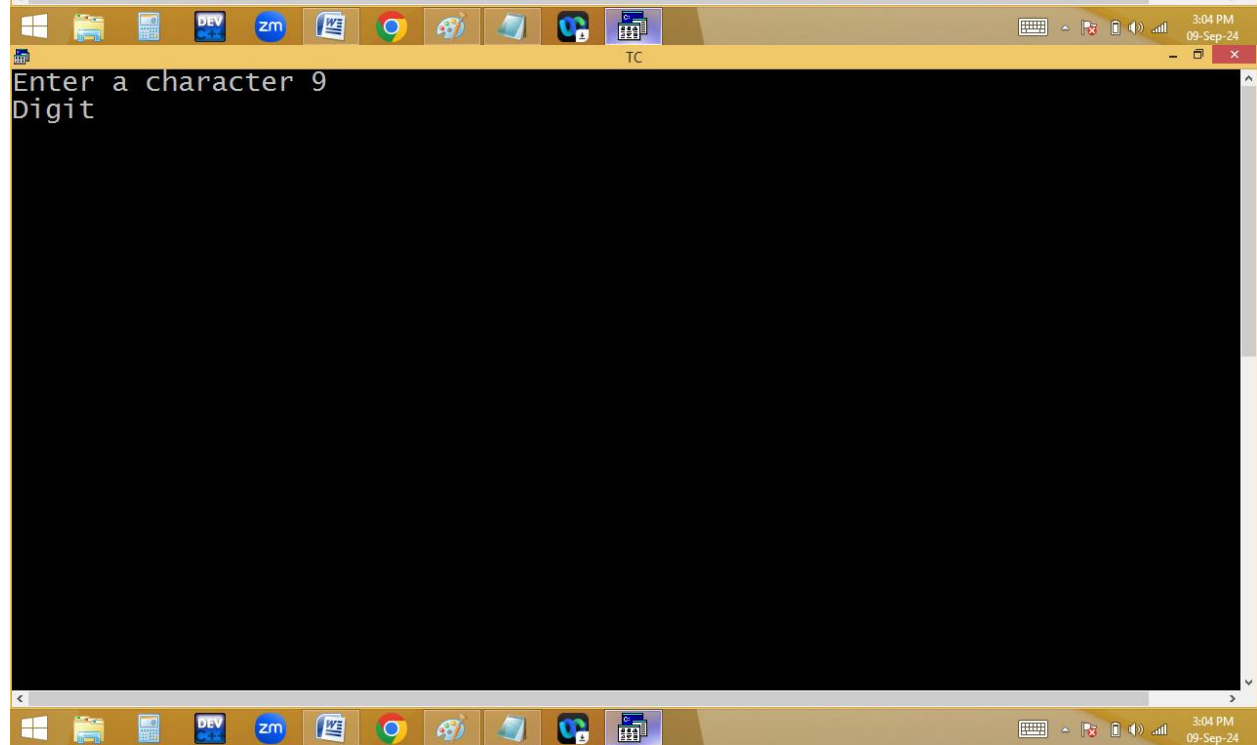
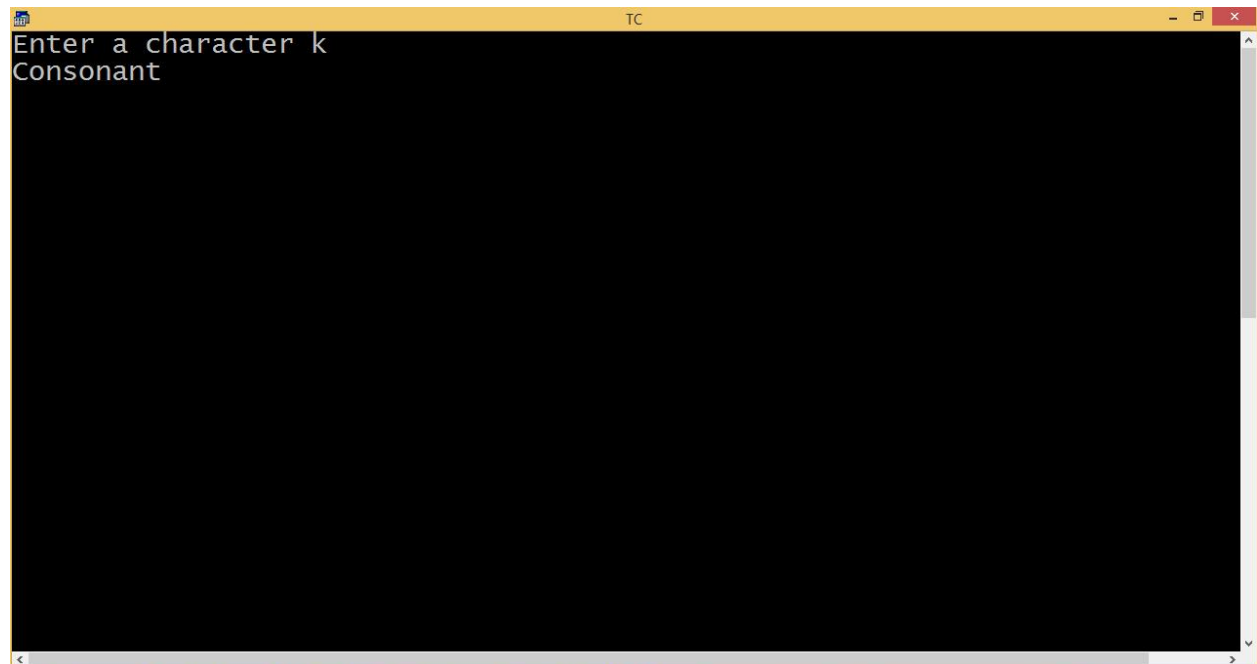


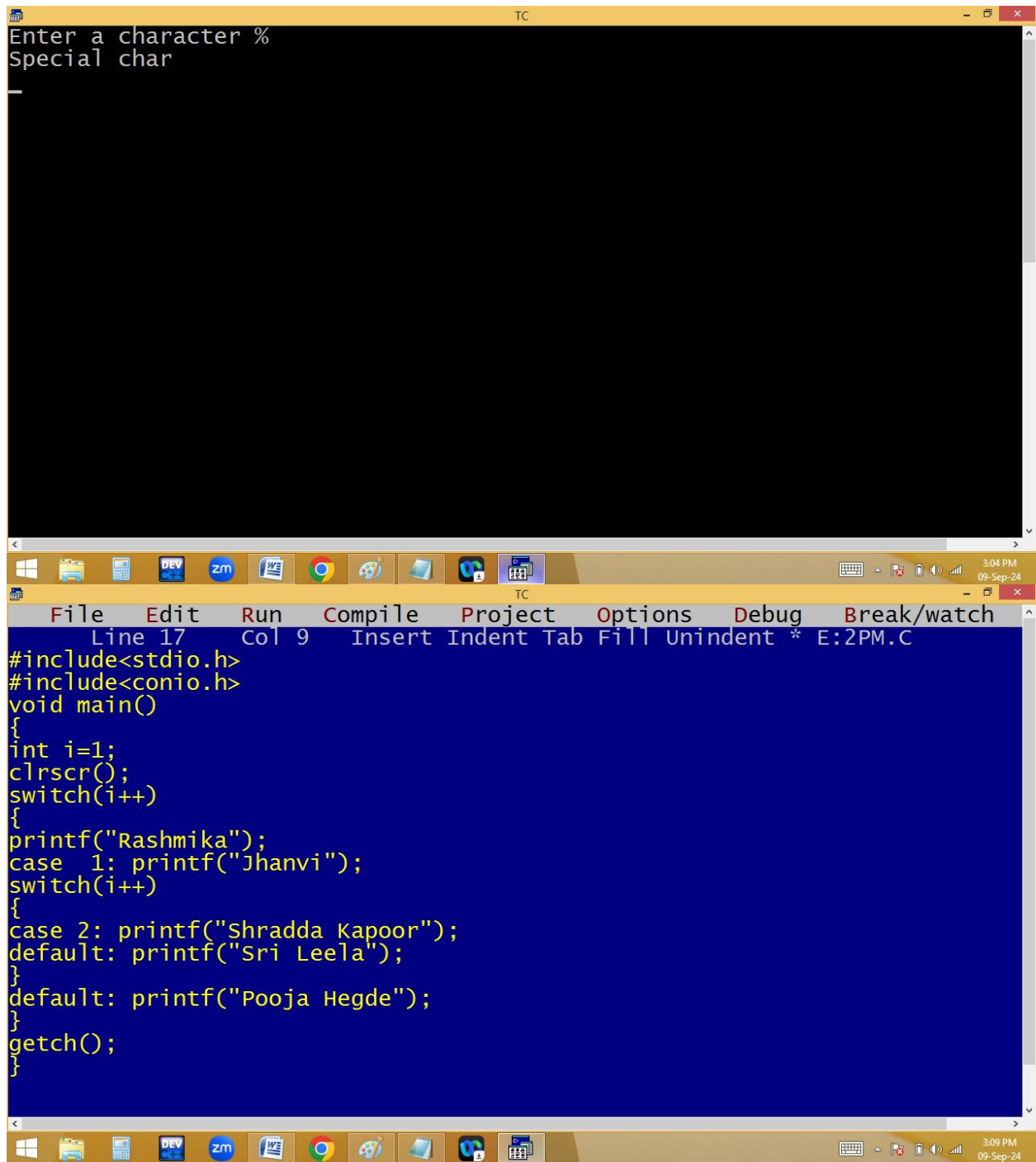
Using predefined functions:



```
File Edit Run Compile Project Options Debug Break/watch
Line 3 Col 18 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
#include<ctype.h>
void main()
{
char ch;
clrscr();
printf("Enter a character ");scanf("%c",&ch);
if(isupper(ch))ch=tolower(ch);
if(isalpha(ch))
{
switch(ch)
{
case 'a': case 'e': case 'i': case 'o': case 'u': puts("Vowel");break;
default: puts("Consonant");break;
}
}
else if(isdigit(ch))puts("Digit");
else puts("Special char");
getch();
}
```

Enter a character o
Vowel





The image shows a screenshot of a Turbo C++ (TC) IDE. The top window is a console with a black background and white text that reads "Enter a character %" and "Special char". Below it is the source code editor with a blue background and yellow text. The code is a C program that uses a switch statement to print names based on input. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the bottom shows the current line and column as "Line 17 Col 9" and the filename as "E:2PM.C". The Windows taskbar at the very bottom shows various application icons and the system clock indicating 3:04 PM on 09-Sep-24.

```
Enter a character %
Special char

#include<stdio.h>
#include<conio.h>
void main()
{
    int i=1;
    clrscr();
    switch(i++)
    {
        printf("Rashmika");
        case 1: printf("Jhanvi");
        switch(i++)
        {
            case 2: printf("Shradda Kapoor");
            default: printf("Sri Leela");
        }
        default: printf("Pooja Hegde");
    }
    getch();
}
```

The image shows a screenshot of a Turbo C++ (TC) IDE. The top window displays the output of a program, which has printed the names "JhanviShradda KapoorSri LeelaPooja Hegde" on a single line. The bottom window shows the source code of the program, which contains a switch statement with a 'continue' statement in the first case. A red error message is displayed at the top of the code window: "Error: Misplaced continue in function main". The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int i=1;
    clrscr();
    switch(i++)
    {
        printf("Rashmika");
        case 1: printf("Jhanvi"); continue;
        switch(i++)
        {
            case 2: printf("Shradda Kapoor");
            default: printf("Sri Leela");
        }
        default: printf("Pooja Hegde");
    }
    getch();
}
/* error */
```

The IDE's menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The Windows taskbar at the bottom shows the time as 3:10 PM on 09-Sep-24.

```
TC
File Edit Run Compile Project Options Debug Break/watch
Line 20 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1;
clrscr();
switch(i++)
{
printf("Rashmika");
case 1: printf("Jhanvi");
switch(i++)
{
case 2: printf("Shradda Kapoor"); break;
default: printf("Sri Leela");
}
default: printf("Pooja Hegde");
}
getch();
}

JhanviShradda KapoorPooja Hegde_
TC
3:11 PM
09-Sep-24
3:12 PM
09-Sep-24
```


The image shows a screenshot of a Turbo C++ (TC) IDE. The top window displays a C program with a switch statement. The code is as follows:

```
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 34 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1;
clrscr();
switch(i++)
{
printf("Rashmika");
case 1: printf("Jhanvi"); break;
switch(i++)
{
case 2: printf("Shradda Kapoor"); break;
default: printf("Sri Leela");
}
default: printf("Pooja Hegde");
}
getch();
}
```

The bottom window shows the output of the program, which is "Jhanvi". The Windows taskbar at the bottom indicates the date as Monday, 9 September, 2024, and the time as 3:13 PM.

Loops / Iterations / Repetitive statements

Loops are used to repeat a block/group of statements continuously until the given condition becomes false.

Loops reduce program size and improves performance.

In loops beginning and ending points are same.

C-Language supports basically 2 types of loops.

- 1. Entry/pre controlled loops.**
- 2. Exit/post controlled loops.**

In entry control loops, condition is tested first and it is true then only statements block is executed.

Under entry control loops we are having

- i. While loop**
- ii. For loop**

In exit control loop, the statements are executed first and later condition is tested.

Under exit control loop we are having

- i. do while.**

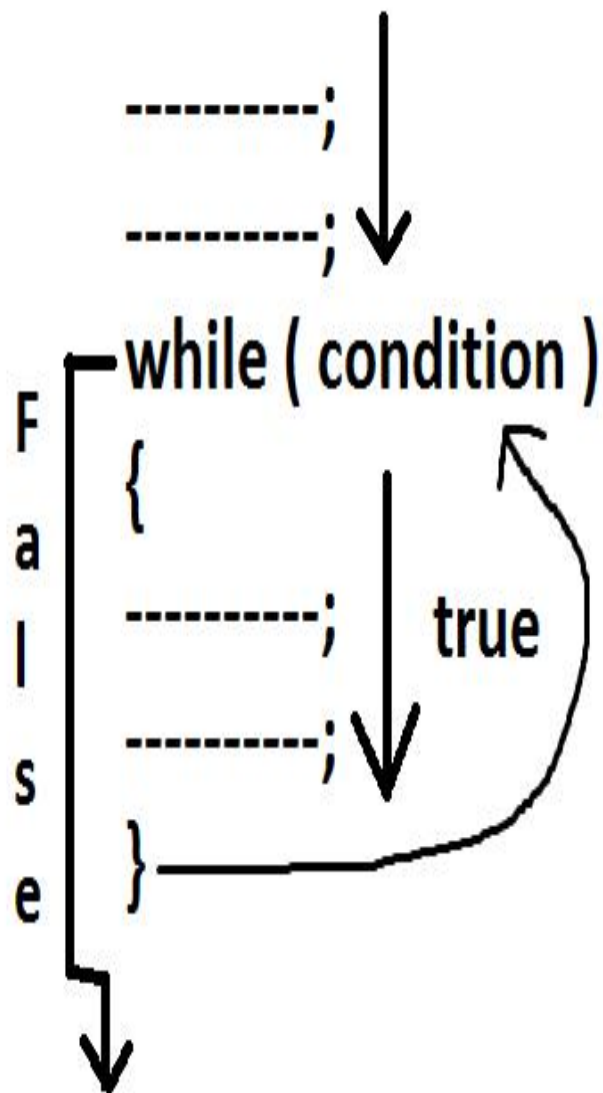
While loop:

- **while is a keyword.**
- **In while loop condition is tested first and it is true then only while block statements are executed. After executing while block statements, the program execution automatically shifted/jumped to while condition at the beginning. If it is true then once again the while block statements are repeated. Like this the**

process is continued until while condition becomes false.

➤ **While is entry control loop.**

Syntax:



Flow chart:

